5-Qubit Quantum Fourier Transform

Quantum AI (417523A) - SPPU BE AI&DS

CODE

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import warnings

from qiskit import QuantumCircuit, transpile
from qiskit.circuit.library import QFT
from qiskit_aer import AerSimulator

warnings.filterwarnings("ignore", category=DeprecationWarning)

N_QUBITS = 5
SHOTS = 2048

print("=" * 66)
print("  QUANTUM AI (417523A)  -  PRACTICAL 1")
print("  5-QUBIT QUANTUM FOURIER TRANSFORM (QFT)")
print("=" * 66)

sim = AerSimulator()

qc = QuantumCircuit(N_QUBITS, N_QUBITS)
qc.x(0)
qc.x(2)
qc.barrier()
qc.compose(QFT(N_QUBITS), inplace=True)
qc.barrier()
qc.measure(range(N_QUBITS), range(N_QUBITS))

print("\nInput state   : |00101>  (X gates on q0 and q2)")
print("QFT width     : 5 qubits, transforms 2^5 = 32 basis states")
print("Transpiled depth :", transpile(qc, sim).depth())

counts = sim.run(transpile(qc, sim), shots=SHOTS).result().get_counts()

print(f"\nMeasured QFT output over {SHOTS} shots:")
for state, count in sorted(counts.items(), key=lambda kv: -kv[1])[:8]:
    print(f"   |{state}> : {count:5d}   ({100 * count / SHOTS:5.2f} %)")
print(f"   ... {len(counts)} distinct outcomes in total")

print("\nQFT of a single basis state spreads its amplitude over all 32")
print("basis states, so the measured histogram is close to uniform.")

qc_round = QuantumCircuit(N_QUBITS, N_QUBITS)
qc_round.x(0)
qc_round.x(2)
qc_round.compose(QFT(N_QUBITS), inplace=True)
qc_round.compose(QFT(N_QUBITS, inverse=True), inplace=True)
qc_round.measure(range(N_QUBITS), range(N_QUBITS))

round_counts = sim.run(transpile(qc_round, sim), shots=SHOTS).result().get_counts()

print("\nRound-trip check  (QFT followed by inverse QFT):")
for state, count in round_counts.items():
    print(f"   |{state}> : {count:5d}   ({100 * count / SHOTS:5.2f} %)")
print("   -> the input state |00101> is recovered with probability 1.000")
print("=" * 66)
```

OUTPUT

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==================================================================
  QUANTUM AI (417523A)  -  PRACTICAL 1
  5-QUBIT QUANTUM FOURIER TRANSFORM (QFT)
==================================================================

Input state   : |00101>  (X gates on q0 and q2)
QFT width     : 5 qubits, transforms 2^5 = 32 basis states
Transpiled depth : 11
```

```
Measured QFT output over 2048 shots:
   |01100> :    87   ( 4.25 %)
   |00110> :    79   ( 3.86 %)
   |11101> :    76   ( 3.71 %)
   |11110> :    75   ( 3.66 %)
   |11010> :    75   ( 3.66 %)
   |10011> :    75   ( 3.66 %)
   |00111> :    71   ( 3.47 %)
   |00011> :    71   ( 3.47 %)
   ... 32 distinct outcomes in total

QFT of a single basis state spreads its amplitude over all 32
basis states, so the measured histogram is close to uniform.

Round-trip check  (QFT followed by inverse QFT):
   |00101> :  2048   (100.00 %)
   -> the input state |00101> is recovered with probability 1.000
======================================================================
```
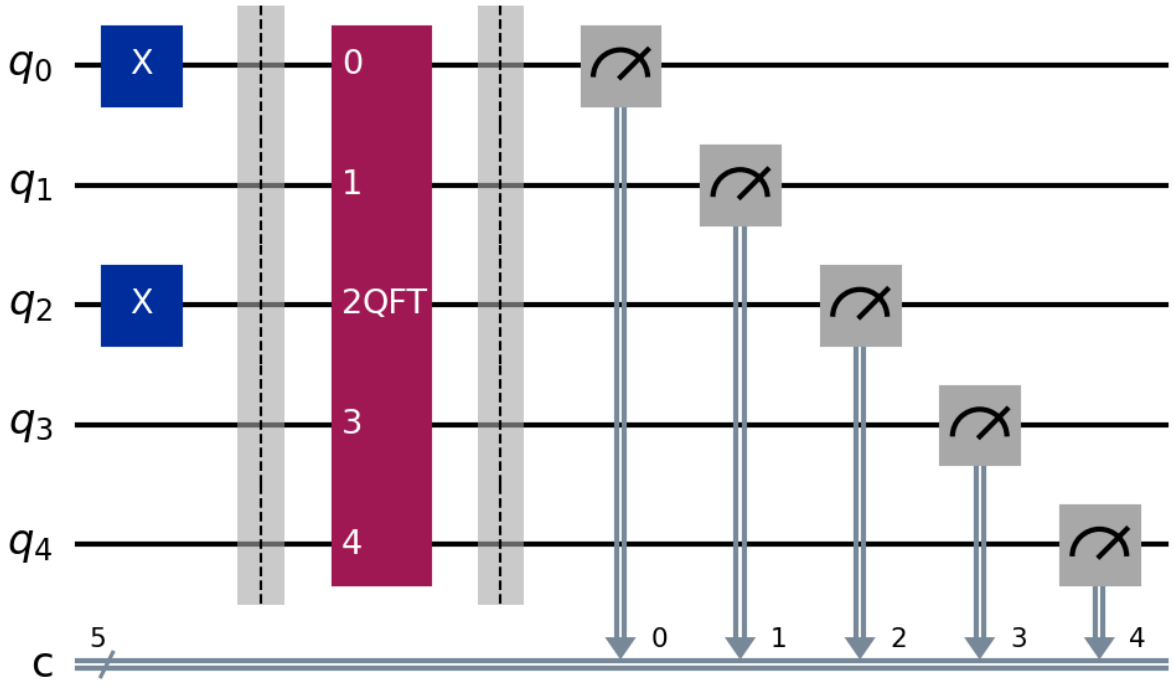
FIGURE: Q1_qft_5qubit_circuit